

第七次作业

干路

$$8.2 t=0^- \text{ 时 } R_{\text{总}} = 5\Omega + 8\Omega + \frac{6\Omega \cdot 3\Omega}{6\Omega + 3\Omega} = 15\Omega \quad I = \frac{U}{R} = 3A$$

$$\therefore U_C = I \cdot 8\Omega = 24V$$

$$6\Omega // 3\Omega \text{ 电阻上电压 } U = I \cdot \frac{6\Omega \cdot 3\Omega}{6\Omega + 3\Omega} = 6V$$

$$\therefore i_L = \frac{U}{3\Omega} = 2A$$

$$\text{由换路定理 } U_C(0^+) = U_C(0^-) = 24V \quad i_L(0^+) = i_L(0^-) = 2A$$

$$\text{换路后, 由 KVL, 有 } U(0^+) + U_C(0^+) - i_L \cdot 8\Omega = 0$$

$$\Rightarrow U(0^+) = -8V$$

$$8.3 \text{ 换路前 } U_1 = \frac{6\Omega}{(9+6)\Omega} \cdot 1.5V = 0.6V$$

$$\text{此时 } U_C(0^-) = 0.6V$$

$$\text{由换路定理 } U_C(0^+) = 0.6V$$

$$\text{换路后, 由 KVL, 有 } i = C \frac{dU_C}{dt}$$

$$\text{由 KVL 有 } U_C + 3\Omega \cdot i + 6\Omega \cdot i - 4i = 0$$

$$\text{即 } U_C + 5C \frac{dU_C}{dt} = 0$$

$$\Rightarrow \frac{dU_C}{dt} - 10U_C = 0 \quad \text{即 } U_C = Ae^{-10t}$$

$$\text{代入 } U_C(0^+) \Rightarrow U_C = (0.6e^{-10t})V$$

$$\therefore i_C = C \frac{dU_C}{dt} = (-0.12e^{-10t})V$$

$$U_1 = i_C \cdot R = (-0.12e^{-10t})V$$

戴维南等效电阻

$$R_d = 6\Omega + \frac{6\Omega \cdot 3\Omega}{6\Omega + 3\Omega} = 8\Omega$$

$$i_L(0^-) = \frac{3\Omega}{3\Omega + 6\Omega} \cdot 9A = 3A$$

由换路定理 $i_L(0^+) = i_L(0^-) = 3A$

稳态时 $i_L(\infty) = 0A$

$$\therefore i_L = i_L(\infty) + (i_L(0^+) - i_L(\infty))e^{-\frac{t}{\tau}} = 3e^{-\frac{t}{\tau}} (A)$$

$$\Rightarrow i_L = 3e^{-2t} (A)$$

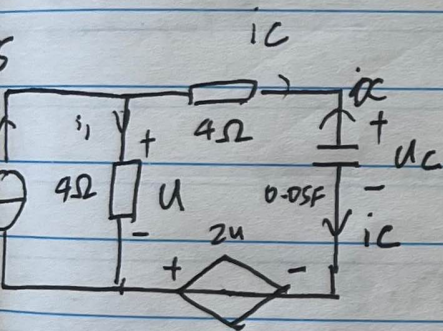
$$u_1 = L \frac{di_L}{dt} = -24e^{-2t} (V)$$

由KVL, 有 $u_1 = i_L \cdot 6\Omega + i_1 \cdot 3\Omega \Rightarrow i_1 = 2e^{-2t} (A)$

$$u_1 = i_1 \cdot 3\Omega = 6e^{-2t} (V)$$

$$P_+ = \frac{1}{2} e^{-4t} \quad P_1 = 12e^{-4t} (W)$$

$$\therefore W = \int_0^{\infty} 12e^{-4t} dt = 3J$$



$u_C(0^+) = u_C(0^-) = 0$. 开关进行开后, 有

$$i_C = C \frac{du_C}{dt}$$

由KCL, $2A = i_1 + i_C \Rightarrow i_1 = (2 - C \frac{du_C}{dt}) A$

由KVL, 有 $3u - 4\Omega \cdot i_C - u_C = 0$

$$u = 4\Omega \cdot i_1$$

$$\Rightarrow 24 - 12C \frac{du_C}{dt} - 4C \frac{du_C}{dt} - u_C = 0$$

$$\Rightarrow 16C \frac{du_C}{dt} + u_C - 24 = 0 \quad \therefore u_C = 24(1 - e^{-1.25t}) V$$

有特解 $u_C = 24V$.

通解为 $0.8 \frac{du_C}{dt} + u_C = 0$

$$\Rightarrow u_C = Ae^{-1.25t}$$

8.7 开关闭合前 $U_C(0^-) = 0$

$$\therefore U_C(0^+) = 0$$

断开后, 有 ~~$U_S - 3\Omega \cdot i - \frac{1}{C} \int i dt = 0$~~

$$\text{断开后有 } U_S - 3\Omega \cdot i - U_C = 0$$

$$3i = C \frac{dU_C}{dt}$$

$$\text{即 } C \frac{dU_C}{dt} + U_C = U_S$$

$$\frac{dU_C}{dt} + U_C = 0 \text{ 特解为 } U_C = A e^{-\frac{t}{\tau}} = A e^{-100t} \text{ (V)}$$

$$\text{稳态时, 有 } U_S - 3\Omega \cdot I - \frac{3I}{j\omega C} = 0$$

$$\Rightarrow I = \frac{U_S}{3-3j}$$

$$\therefore U_{PC} = \frac{3I}{j\omega C} = 5-5j \text{ V}$$

特解

$$\text{即 } U_{PC} = 10 \cos(100t - 45^\circ) \text{ (V)}$$

$$\therefore U_C = A e^{-100t} + 10 \cos(100t - 45^\circ) \text{ (V)}$$

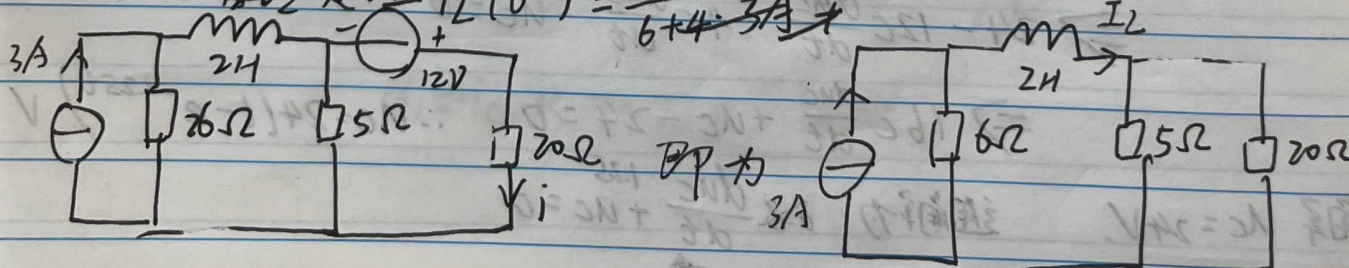
$$\text{代入 } U_C(0^+) = 0 \Rightarrow A = -5\sqrt{2}$$

$$\Rightarrow U_C = -5\sqrt{2} e^{-100t} + 10 \cos(100t - 45^\circ) \text{ V}$$

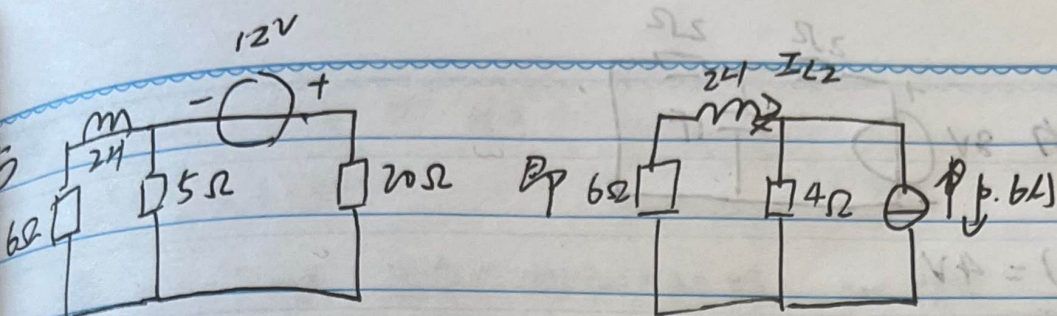
8.9 对电容: 开关闭合前, $5\Omega // 20\Omega = \frac{5 \times 20}{5+20} = 4\Omega$

由戴维南定理

$$U_{OC} = \frac{6}{6+4} \cdot 3A = 1.8V$$



$$I_{L1} = \frac{6}{6+4} \cdot 3A = 1.8A$$



$$I_{L2} = \frac{4}{6+4} \cdot 0.6 = 0.24 \text{ A}$$

$$I_L(10^-) = I_{L1} + I_{L2} = 2.04 \text{ A}$$

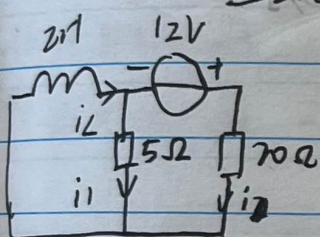
$$I_L(10^+) = 2.04 \text{ A}$$

$$I_L(\infty) = \frac{12}{20} \text{ A} = 0.6 \text{ A}$$

$$R_{th} = 5\Omega // 20\Omega = 4\Omega \quad \tau = \frac{L}{R} = 0.5 \text{ s}$$

$$i_L = i_L(\infty) + (i_L(10^-) - i_L(\infty))e^{-\frac{t}{\tau}}$$

$$= 2.04e^{-2t} \text{ (A)} = 0.6 + 1.44e^{-2t} \text{ (A)}$$



由 KCL 有 $i_L = i_1 + i_2$

由 KVL 有 $12V - 20\Omega \cdot i_2 + 5\Omega \cdot i_1 = 0$

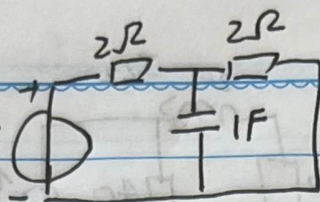
$$\Rightarrow i_2 = 0.48 + 0.408e^{-2t} \text{ (A)}$$

$$u_L = L \frac{di_L}{dt} = -4.08e^{-2t} \text{ (V)}$$

$$\therefore \text{有 } 12V = u_L + 20 \cdot i \Rightarrow i = 0.6 + 0.288e^{-2t} \text{ (A)}$$

$t \rightarrow \infty$ 时

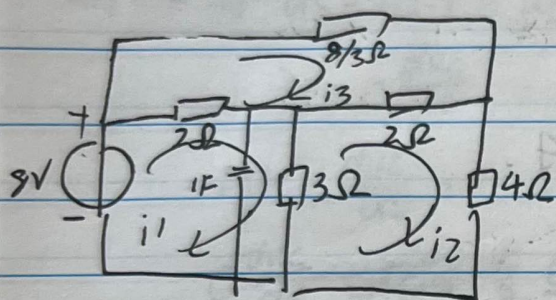
8.10 $t \rightarrow \infty$ 时, 电路等效为 8V



$$U_C(0^+) = U_C(0^-) = 4V$$

0.

~~$t \rightarrow \infty$~~ $t \rightarrow \infty$ 时,



由回路电流法

$$5\Omega \cdot i_1 - 3\Omega \cdot i_2 - 2\Omega \cdot i_3 = 8V$$

$$9\Omega \cdot i_1$$

$$-3\Omega \cdot i_1 + 9\Omega \cdot i_2 - 2\Omega \cdot i_3 = 0$$

$$-2\Omega i_1 - 2\Omega i_2 + (4 + \frac{8}{3})\Omega i_3 = 0$$

$$\Rightarrow i_1 = 2.8A$$

$$i_2 = 1.2A$$

$$i_3 = 1.2A$$

$$\therefore 3\Omega \text{ 电阻上电流 } i = i_1 - i_2 = 1.6A$$

$$U_C(0^+) = iR = 4.8V = U_C(0^+)$$

求等效电阻

$$R = \frac{2\Omega // [(2\Omega // 3\Omega) + (4\Omega // 2\Omega)]}{1} = \frac{7}{8}\Omega$$

$$R = 1\Omega$$

$$\tau = RC = 1s$$

$$t = RC = \frac{7}{8}s$$

$$\therefore U_C = U_C(\infty) + (U_C(0^+) - U_C(\infty))e^{-\frac{t}{\tau}}$$

$$= 4.8 - 0.8e^{-\frac{6t}{7}} (V)$$

$$U_C = U_C(\infty) + (U_C(0^+) - U_C(\infty))e^{-\frac{t}{\tau}}$$

$$= 4 + 0.8e^{-t} (V)$$

$$15 \quad \tau = RC = 0.5S \Rightarrow R = 50\Omega$$

$$\therefore \tau' = \frac{L}{R} = 0.1S$$

$t=0^+$ 时, 电容C相当于短路

$$u=5V$$

$t \rightarrow \infty$ 时 电容C相当于断路

$$u=10V$$

$t=0^+$ 时 电感L相当于断路

$$u=10V$$

$$u(0^+) = 10V \quad u=10V$$

$t \rightarrow \infty$ 时 电感L相当于短路

$$u=5V$$

$$u=5V$$

$$\therefore u = u(\infty) + (u(0^+) - u(\infty))e^{-\frac{t}{\tau}}$$

$$= 15 + 5e^{-10t} \quad V$$

$$\therefore u = 15 + 5e^{-10t} \quad V$$

由KCL, 有 $i_s = i + i_L$

$$\text{由KVL, 有 } 2i + L \frac{di_L}{dt} + 3i = 0$$

$$\Rightarrow L \frac{di_L}{dt} - 5i + 3i_L = 0$$

$$\text{即 } L \frac{di_L}{dt} + 3i_L = 5i$$

$$\text{通解为 } i_L = Ae^{-\frac{5}{L}t} = Ae^{-50t}$$

设特解为

$$i_L = A_1 e^{-10t} + A_2 t e^{-10t}$$

$$\text{代入 } i_L(0^+) = 3A$$

$$\Rightarrow A_1 = 3$$

$$\text{稳态时, 有 } 2i - 3i = 0 \Rightarrow i = 0$$

$$i_L = i_s$$

$$\therefore \text{解为 } (Ae^{-\frac{10}{L}t} + 5e^{-10t}) A$$

$$\text{代入 } i_L(0^+) = 3A \Rightarrow i_L = -2e^{-2t} + 5e^{-10t} A$$

$$\therefore i = i_s - i_L = 2e^{-2t} A$$

强制分量 0

自由分量 $2e^{-2t} A$

$$\text{解为 } 3e^{-10t} + 50te^{-10t}$$

$$\therefore i = i_s - i_L = 2e^{-2t} - (3e^{-10t} + 50te^{-10t}) A$$

代入原式有

$$-30e^{-10t} + A_2 e^{-10t} + 5A_2 t e^{-10t} + 5A_1 e^{-10t} + A_2 t e^{-10t} = 25e^{-10t}$$

$$\Rightarrow -15 + 0.5A_2 + 5A_1 = 25$$

$$\Rightarrow A_2 = 50$$

$$8.17 \quad t=0^- \text{ 时 } U_C = \frac{4R^2}{(12+4)\Omega} \cdot 32V = 8V$$

$$\therefore U_C(0^+) = U_C(0^-) = 8V$$

$$\text{换路后, 有 } U_S - C \frac{dU_C}{dt} \cdot 10\Omega - U_C = 0$$

$$\Rightarrow 0.1 \frac{dU_C}{dt} + U_C = 10t$$

$$\text{通解为 } U_C = Ae^{-10t}$$

$$\text{特解形式为 } Bt + C, \text{ 代入有 } 0.1B + Bt + C = 10t$$

$$\Rightarrow B=10 \quad C=-1$$

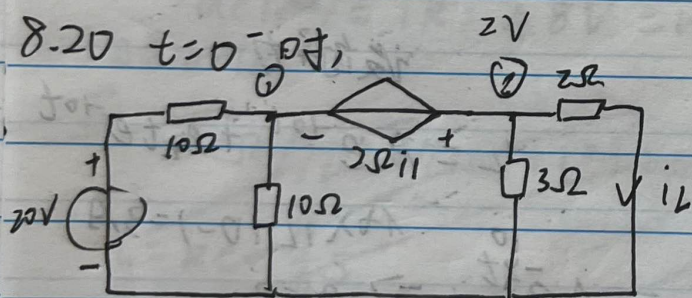
$$\therefore \text{解为 } U_C = Ae^{-10t} + 10t - 1$$

$$\text{代入 } U_C(0^+) = 8V \Rightarrow A=9$$

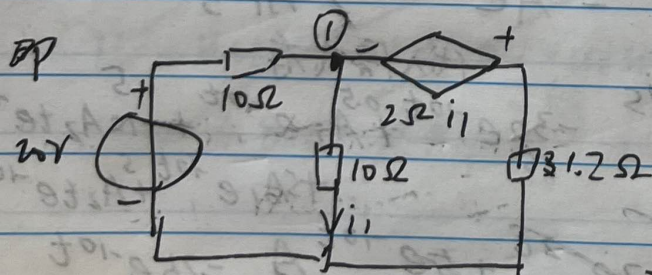
$$\therefore U_C = (9e^{-10t} + 10t - 1)V$$

$$\text{自由分量 } 9e^{-10t}V \quad \text{强制分量 } 10t - 1 \quad \checkmark$$

$$8.20 \quad t=0^- \text{ 时,}$$



由结点电压法



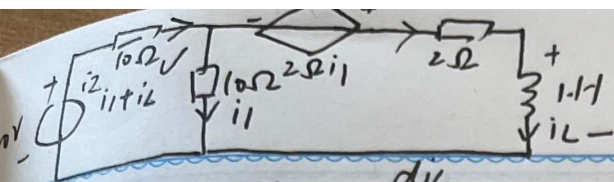
$$\left(\frac{1}{10} + \frac{1}{10} + \frac{1}{1.2}\right) U_1 = \frac{20V}{10\Omega} - \frac{2\Omega i_1}{1.2\Omega}$$

$$U_1 = 10\Omega \cdot i_1$$

$$\Rightarrow U_1 = \frac{5}{3}V \quad i_1 = \frac{1}{6}A$$

$$\therefore U_2 = U_1 + 2\Omega i_1 = 2V$$

$$i_2 = \frac{U_2}{2\Omega} = 1A$$



开关断开后, $U_L = L \frac{di_L}{dt}$ 由KCL, $i_2 = i_1 + i_L$

由KVL, 有 $20V - 10\Omega(i_1 + i_L) - 10\Omega i_1 = 0$

$$20V = 10\Omega i_1 + 10\Omega(i_1 + i_L)$$

$$2\Omega i_1 - 2\Omega \cdot i_L + 10\Omega i_1 - L \frac{di_L}{dt} = 0$$

$$\text{即有 } L \frac{di_L}{dt} + 8i_L = 12$$

通解
特解为 $i_L = A e^{-8t}$

特解为 $i_L = 1.5A$ $\therefore$ 解为 $(A e^{-8t} + 1.5)A$

$$\text{代入 } i_L(0^+) = 1A \Rightarrow A = -0.5$$

$$\therefore t > 0 \text{ 时 } i_L = (-0.5 e^{-8t} + 1.5)A$$

$$I_{PL} = \frac{U_{S1}}{R + jX_L} = 3 \angle 0^\circ$$

22 稳态时

$t=0^-$ 时, 由叠加定理 对 $U_{S1}(t)$ $X_L = \omega L = 10\Omega$

$$X_C = \frac{1}{\omega C} = 10\Omega$$

$$X_L // X_C // R = 5\Omega$$

$$\therefore U_C = \frac{5\Omega}{10+5}\Omega U_{S1} = 10\sqrt{2} \cos(100t + 45^\circ) V$$

$$i_L = \frac{U_C}{X_L} = \sqrt{2} \cos(100t - 45^\circ) V$$

$$i_L = \frac{1}{L} \int U_C dt$$

$$\text{由 } U_C = L \frac{di_L}{dt} \Rightarrow i_L = \sqrt{2} \cos(100t - 45^\circ) V$$

对 U_{S2} $i_L = \frac{U_{S2}}{5\Omega} = 4A$ $U_C = 0$

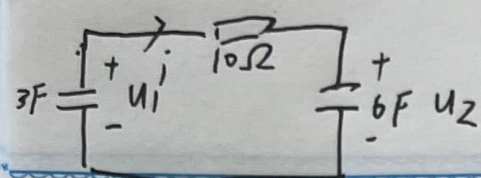
$$\therefore i_L(0^-) = i_L(0^+) = 4\sqrt{2} \cos(100t - 45^\circ) V \quad 4 + \sqrt{2} \cdot \frac{\sqrt{2}}{2} = 5A$$

$$U_C(0^-) = U_C(0^+) = 10\sqrt{2} \cos(100t + 45^\circ) V = 10\sqrt{2} \cdot \frac{\sqrt{2}}{2} = 10V$$

$$T_2 = RC = 5 \times 10^{-3} s$$

$$\text{稳态时 } U_C(\infty) = U_{S2} = 20V$$

$$\therefore U_C = U_C(\infty) + (U_C(0^+) - U_C(\infty)) e^{-\frac{t}{T_2}} = 20 - 10 e^{-200t} (V)$$



$$8.28 \quad i = -C_1 \frac{du_1}{dt} = -C_2 \frac{du_2}{dt}$$

$$\text{由 KVL, } u_1 - 10i - u_2 = 0$$

$$\therefore u_1 = 10i + u_2$$

$$\therefore -\frac{du_1}{dt} = 10 \frac{di}{dt} - \frac{du_2}{dt}$$

$$\Rightarrow -\frac{i}{3} = 10 \frac{di}{dt} + \left(10 - \frac{1}{6}\right) i \Rightarrow -i = 20 \frac{di}{dt} \Rightarrow i = Ae^{-\frac{t}{20}}$$

$$u_1 = \int \frac{i}{C_1} dt = \frac{20}{3} e^{-\frac{t}{20}} + C$$

$$u_2 = \int \frac{i}{C_2} dt = \frac{10}{3} e^{-\frac{t}{20}} + C$$

$$\therefore u_1 = \int_0^t \frac{i}{C_1} dt = \left(\frac{20A}{3} e^{-\frac{t}{20}} - \frac{20A}{3} + 20 \right) V$$

$$u_2 = \int_0^t \frac{i}{C_2} dt = \left(\frac{10A}{3} - \frac{10A}{3} e^{-\frac{t}{20}} \right) V$$

$$\therefore \text{当 } t \rightarrow \infty \text{ 时 } u_1 = u_2, \therefore -\frac{20A}{3} + 20 = \frac{10A}{3}$$

$$\Rightarrow A = 2$$

$$\therefore u_1 = \left(\frac{40}{3} e^{-\frac{t}{20}} + \frac{20}{3} \right) V \quad u_2 = \left(\frac{20}{3} - \frac{20}{3} e^{-\frac{t}{20}} \right) V$$

8.30 由 KVL, 有当 $0 < t < 1s$ 时.

$$u_s - L \frac{di_L}{dt} - 10\Omega(i_L + i_s) = 0$$

$$\Rightarrow L \frac{di_L}{dt} + 10\Omega i_L = u_s - 10\Omega i_s$$

$$t=0 \text{ 时 } i_L = 0$$

$$\text{即 } L \frac{di_L}{dt} + 10\Omega i_L = \varepsilon(t) = 10V$$

$$\text{通解为 } i_L = Ae^{-5t} \quad \text{特解为 } i_L = 0.1A$$

$$\therefore \text{解为 } i_L = 0.1 - 0.1e^{-5t} A$$

$$\therefore i_L(1-) = i_L(1+) = (0.1 - 0.1e^{-5}) A$$

在 $t=1s$ 时

$$\text{有 } u_s - L \frac{di_L}{dt} - 10\Omega i_L + i_s = 0$$

$$\text{即 } L \frac{di_L}{dt} + 10\Omega i_L = \varepsilon(t) - 10\varepsilon(t-1)$$

$$\text{通解为 } i_L = Ae^{-5t} \quad \text{特解为 } i_L = -0.9A$$

$$\text{代入 } i_L(1-) = (0.1 - 0.1e^{-5}) A$$

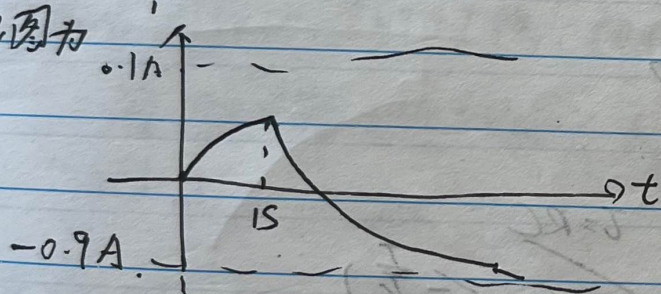
$$\Rightarrow A = e^5 - 0.1$$

$$\therefore i_L = (e^5 - 0.1)e^{-5t} - 0.9A$$

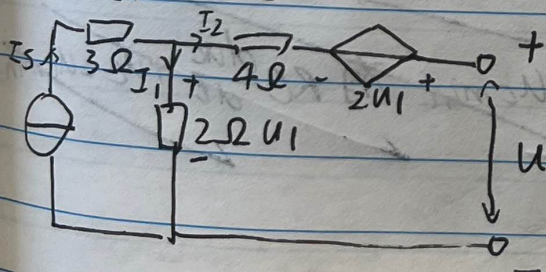
$$= e^{5-5t} - 0.1e^{-5t} - 0.9A$$

$$\therefore i_L = [0.1 - 0.1e^{-5t}] \varepsilon(t) + [e^{5-5t} - 1] \varepsilon(t-1) A$$

波形图



8.3 对左



$$\text{开路时 } u_1 = I_5 \cdot 2\Omega = 2V$$

$$\therefore \text{开路电压 } u = u_1 + 2u_1 = 6V$$

$$\text{短路时, 由 KVL, 有 } u_1 + 2u_1 - 4\Omega \cdot I_2 = 0$$

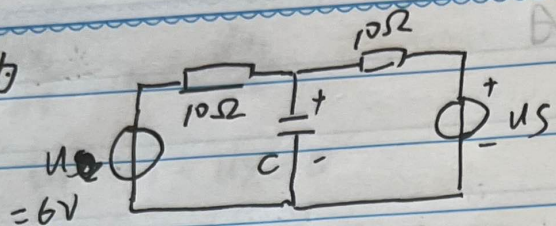
$$u_1 = I_1 \cdot 2\Omega$$

$$I_5 = I_1 + I_2$$

$$\Rightarrow I_2 = 0.6A \quad I_1 = 0.4A$$

$$\therefore \text{戴维南等效电阻为 } \frac{u}{I_2} = 10\Omega$$

原电路等效为



$$X_C = \frac{1}{\omega C} = 5\Omega$$

$$10\Omega // -jX_C = (2-4j)\Omega$$

由叠加定理 $U_C(0^-) = \frac{10\Omega}{20\Omega} \cdot u_s + \frac{2-4j}{12-4j} \cdot u_s$
 $= (8-5j)V = 9.43 \angle -32.0^\circ V$

$\therefore U_C(0^-) = 9.43 \angle -32.0^\circ V$
 $9.43 \cos(10t - 32.0^\circ) V = 8V$

稳态时 $U_C(\infty) = 6V$

$\therefore U_C(t) = U_C(\infty) + [U_C(0^-) - U_C(\infty)]e^{-\frac{t}{\tau}}$ $\tau = RC = 0.2s$
 $= 6 + 2e^{-5t} (V)$

8-32 0~T 时, 为零状态响应, 初值为 0

有 $RC \frac{dU_C}{dt} + U_C = U_S$
 $\Rightarrow U_C = U_S(1 - e^{-\frac{t}{\tau}})$ $\tau = RC$

T~2T 时, 为零输入响应, 初值为 $U_S(1 - e^{-\frac{T}{\tau}})$

存在 ZI 响应! $U_C = U_S(1 - e^{-\frac{T}{\tau}}) \cdot e^{-\frac{t-T}{\tau}}$ 平移反形, 故

8-32 稳态时 0~T, 为零状态响应, 初值为 0

$\Rightarrow U_C = U_{Cmin}(1 - e^{-\frac{t}{\tau}})$ $\tau = RC$

在 T~2T 时, 为零输入响应

8.32. 稳态时, 在 $2KT \sim 2KT+T$ 段, 初值为 U_{Cmin} , 为零状态响应

$$有 RC \frac{dU_C}{dt} + U_C = U_{Cmin}$$

$$\Rightarrow U_{C1} = U_{Cmin} (1 - e^{-\frac{(t-2KT)}{\tau}}) \quad \tau = RC$$

$2KT+T \sim 2KT+2T$ 段, 初值为 U_{Cmax} , 为零输入响应

$$U_{C2} = U_{Cmax} e^{-\frac{(t-2KT-T)}{\tau}} \quad \tau = RC$$

$$\because U_{C1}(2KT+T) = U_{Cmax}$$

$$\Rightarrow U_{Cmax} = U_{Cmin} (1 - e^{-\frac{T}{\tau}})$$

$$U_{C2}(2KT+2T) = U_{min}$$

$$\Rightarrow U_{min} = U_{Cmax} e^{-\frac{T}{\tau}} \quad \text{即有}$$

8.32. 稳态时 在 $2KT \sim 2KT+T$ 段, 初值为 U_{Cmin} , 为零状态响应

$$有 RC \frac{dU_C}{dt} + U_C = U_{Cmin} + U_S$$

$$\Rightarrow U_{C1} = (U_{Cmin} - U_S) e^{-\frac{(t-2KT)}{RC\tau}} + U_S \quad \tau = RC$$

在 $2KT+T \sim 2KT+2T$ 段, 初值为 U_{Cmax} , 为零输入响应

$$有 RC \frac{dU_C}{dt} + U_C = 0$$

$$\Rightarrow U_{C2} = U_{Cmax} e^{-\frac{(t-2KT-T)}{\tau}} \quad \tau = RC$$

$$\because U_{C1}(2KT+T) = U_{Cmax}$$

$$U_{C2}(2KT+T) = U_{Cmin}$$

$$\Rightarrow (U_{Cmin} - U_S) e^{-\frac{T}{\tau}} + U_S = U_{Cmax}$$

$$U_{Cmax} e^{-\frac{T}{\tau}} = U_{Cmin}$$

$$U_{Cmin} = \frac{U_S}{e^{-\frac{T}{\tau}} + 1}$$

解得 $U_{Cmin} = \frac{U_S (e^{-\frac{T}{\tau}} - 1)}{(1 - e^{-\frac{T}{\tau}} - e^{-\frac{T}{\tau}})} = \frac{U_S (e^{-\frac{T}{\tau}} - 1)}{e^{-\frac{T}{\tau}} (1 - e^{-\frac{T}{\tau}} - 1) (e^{-\frac{T}{\tau}} + 1)} = \frac{U_S e^{-\frac{T}{\tau}}}{e^{-\frac{T}{\tau}} + 1} \quad \tau = RC$

$$8.35 \quad i_C = C \frac{du_C}{dt}$$

有 $i = 3i + i_C \Rightarrow i_C = -2i$ $t > 0^+$ 时

由 KVL, 有 $u_S - 8i + 6i - u_C = 0$ $(u_S = 0)$

$$\Rightarrow u_C - 0.02 \frac{du_C}{dt} = u_S = 0$$

通解为 $u_C = Ae^{50t}$

~~稳态时 $u_S = 0 = u_C$~~ $u_C = Ae^{50t}$ $u_S = A$

为求特解, 两边从 0^- 到 0^+ 积分, 有

$$-0.02 u_C(0^+) + 0.02 u_C(0^-) = 1$$

且有

$$\Rightarrow u_C(0^+) - u_C(0^-) = -50V \quad u_C(0^-) = 0$$

代入 $u_C(0^+) = -50V$ $\therefore t > 0$ 时

得 $u_C = -50e^{50t} (V)$

$\therefore u_C = -50e^{50t} \varepsilon(t) (V)$

